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Lab task : 8

Question 1:

Code :

```
lab_tasks > labtask8 > C++ Q1.cpp > ...
1  #include<iostream>
2  using namespace std;
3
4  class bankAccount
5  {
6      private :
7          static int totalBalance;
8
9      public:
10
11      bankAccount()
12      {
13          int deposit= 0;
14          totalBalance += deposit ;
15      }
16
17      bankAccount(int deposit)
18      {
19          totalBalance += deposit;
20      }
21
22      ~bankAccount()
23      {
24          cout<<"Account closed!\n";
25      }
26
27      static void display()
28      {
29          cout<<"Current total balance : "<<totalBalance<<endl;
30      }
31  };
32
33
34
35  int bankAccount::totalBalance = 0;
36
37  int main()
38  {
39      bankAccount A1;
40      A1.display();
41
42      bankAccount A2(45);
43      A2.display();
44
45      bankAccount A3(20);
46      A3.display();
47
48  }
```

Output:

```
Current total balance : 0  
Current total balance : 45  
Current total balance : 65  
Account closed!  
Account closed!  
Account closed!  
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```

Question 2:

Code :

```
1 // libbook.h : <libbook>
2
3 #include<iostream>
4 #include <string>
5 using namespace std;
6
7 class LibraryBook
8 {
9     private :
10     static int maxBooks;
11     static int bookCount;
12     string name;
13     int ID;
14
15     public :
16
17     LibraryBook() : name("unknown"), ID(0)
18     {
19         if(bookCount< maxBooks)
20         {
21             bookCount++;
22         }
23         else
24         {
25             cout<<"This library is full!! \nYou cannot add more books\n";
26         }
27     }
28
29     LibraryBook(string name, int ID)
30     {
31         if(bookCount<= maxBooks)
32         {
33             this->name = name;
34             this->ID = ID;
35             bookCount++;
36         }
37         else
38         {
39             cout<<"This library is full!! \nYou cannot add more books\n";
40         }
41     }
42
43     ~LibraryBook()
44     {
45         bookCount--;
46         cout<<"Book removed or returned\n";
47     }
48 }
```

```
47
48     static void display()
49     {
50         cout<<"Current count of books : "<<bookCount<<endl;
51     }
52 }
53
54
55 };
56
57 int LibraryBook::maxBooks = 5;
58 int LibraryBook::bookCount = 0;
59
60 int main()
61 {
62     LibraryBook b1;
63     b1.display();
64
65     LibraryBook b2("nanhay doggy ",4562);
66     b2.display();
67
68     LibraryBook b3;
69     b3.display();
70
71     LibraryBook b4("Lakarhara", 784);
72     b4.display();
73
74     LibraryBook b5;
75     b5.display();
76
77
78 }
```

Output:

Current count of books : 1

Current count of books : 2

Current count of books : 3

Current count of books : 4

Current count of books : 5

Book removed or returned

Book removed or returned

Book removed or returned

Book removed or returned

Book removed or returned

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Question 3:

Header file:

```
1  #include<iostream>
2  #include <string>
3  using namespace std;
4
5  class Employee
6  {
7      private :
8          static int maxlimit;
9          static int employeeCount;
10         string name;
11         int ID;
12
13     public :
14
15         Employee();
16
17
18         Employee(string name, int ID);
19
20         ~Employee();
21
22         static void display();
23
24
25 };
26
```

Cpp file:


```
1  #include <iostream>
2  #include "Q3.h"
3  using namespace std;
4
5  int Employee::maxlimit = 5;
6  int Employee::employeeCount = 0;
7
8  Employee::Employee() : name("unknown"), ID(0)
9  {
10     if (employeeCount < maxlimit)
11     {
12         employeeCount++;
13     }
14     else
15     {
16         cout << "This library is full!! \nYou cannot add more books\n";
17     }
18 }
19
20 Employee::Employee(string name, int ID)
21 {
22     if (employeeCount < maxlimit)
23     {
24         this->name = name;
25         this->ID = ID;
26         employeeCount++;
27     }
28     else
29     {
30         cout << "No more employees can be added! \n";
31     }
32 }
33
34 Employee::~Employee()
35 {
36     employeeCount--;
37     cout << "employee removed \n";
38 }
39
40 void Employee::display()
41 {
42     cout << "Current number of employees : " << employeeCount << endl;
43 }
44
45
```

Main cpp file:

```
lab_tasks > labtask8 > Q33 > main.cpp > ...
1  #include<iostream>
2  #include"Q3.h"
3  using namespace std;
4
5
6  int main()
7  {
8      Employee E1;
9      E1.display();
10
11     Employee E2("Ali ", 5678);
12     E2.display();
13
14     Employee E3;
15     E3.display();
16
17     Employee E4("Wali", 4564);
18     E4.display();
19
20     Employee E5;
21     E5.display();
22 }
```

Output:

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Current number of employees : 1

Current number of employees : 2

Current number of employees : 3

Current number of employees : 4

Current number of employees : 5

employee removed

employee removed

employee removed

employee removed

employee removed

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